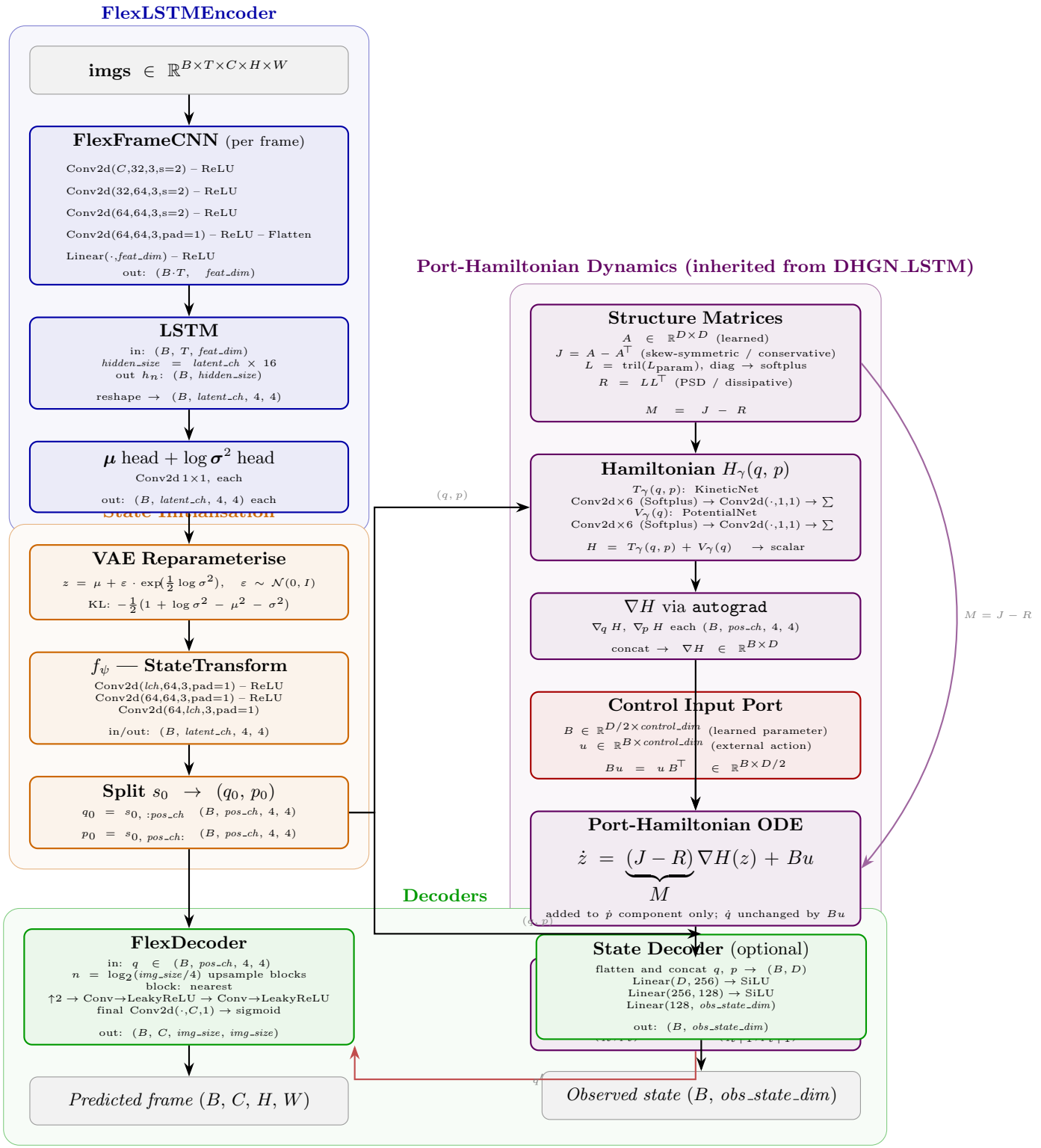
FlexLSTMEncoder
imgs ∈ $\mathbb{R}^{B\times T\times C\times H\times W}$
FlexFrameCNN (per frame)
Conv2d(C,32,3,s=2) – ReLU
Conv2d(32,64,3,s=2) – ReLU
Conv2d(64,64,3,s=2) – ReLU
Conv2d(64,64,3,pad=1) – ReLU – Flatten
Linear(·,feat_dim) – ReLU
out: (B·T, feat_dim)
LSTM
in: (B, T, feat_dim)
hidden_size = latent_ch × 16
out h_n: (B, hidden_size)
reshape → (B, latent_ch, 4, 4)
μ head + $\log\sigma^2$ head
Conv2d 1×1, each
out: (B, latent_ch, 4, 4) each
State initialisation
VAE Reparameterise
$z = \mu + \varepsilon \cdot \exp(\frac{1}{2}\log\sigma^2)$, $\varepsilon \sim \mathcal{N}(0, I)$
KL: $-\frac{1}{2}(1 + \log\sigma^2 - \mu^2 - \sigma^2)$
f_ψ — StateTransform
Conv2d(lch,64,3,pad=1) – ReLU
Conv2d(64,64,3,pad=1) – ReLU
Conv2d(64,lch,3,pad=1)
in/out: (B, latent_ch, 4, 4)
Split s_0 → (q_0, p_0)
$q_0 = s_{0,\,:pos_ch}$ (B, pos_ch, 4, 4)
$p_0 = s_{0,\,pos_ch:}$ (B, pos_ch, 4, 4)
Port-Hamiltonian Dynamics (inherited from DHGN_LSTM)
Structure Matrices
$A \in \mathbb{R}^{D\times D}$ (learned)
$J = A - A^\top$ (skew-symmetric / conservative)
$L = \mathrm{tril}(L_{\mathrm{param}})$, diag → softplus
$R = LL^\top$ (PSD / dissipative)
$M = J - R$
Hamiltonian $H_\gamma(q, p)$
$T_\gamma(q, p)$: KineticNet
Conv2d×6 (Softplus) → Conv2d(·,1,1) → $\sum$
$V_\gamma(q)$: PotentialNet
Conv2d×6 (Softplus) → Conv2d(·,1,1) → $\sum$
$H = T_\gamma(q, p) + V_\gamma(q)$ → scalar
∇H via autograd
$\nabla_q H$, $\nabla_p H$ each (B, pos_ch, 4, 4)
concat → $\nabla H \in \mathbb{R}^{B\times D}$
Control Input Port
$B \in \mathbb{R}^{D/2\times control_dim}$ (learned parameter)
$u \in \mathbb{R}^{B\times control_dim}$ (external action)
$Bu = u\,B^\top \in \mathbb{R}^{B\times D/2}$
Port-Hamiltonian ODE
$\dot z = \underbrace{(J-R)}_{M}\nabla H(z) + Bu$
added to $\dot p$ component only; $\dot q$ unchanged by Bu
(q, p)
$M = J - R$
Decoders
FlexDecoder
in: $q \in$ (B, pos_ch, 4, 4)
$n = \log_2(img_size/4)$ upsample blocks
block: nearest
↑2 → Conv→LeakyReLU → Conv→LeakyReLU
final Conv2d(·,C,1) → sigmoid
out: (B, C, img_size, img_size)
State Decoder (optional)
flatten and concat q, p → (B, D)
Linear(D, 256) → SiLU
Linear(256, 128) → SiLU
Linear(128, obs_state_dim)
out: (B, obs_state_dim)
Predicted frame (B, C, H, W)
Observed state (B, obs_state_dim)